

Element I

The objective of this testing procedure was to evaluate the design through a series of performance and durability tests of the cart handle. The idea was focused around removability, ease of use, and ergonomics which were all previously provided by the client as some current issues with the design.

The procedure was run several times with multiple different devices to evaluate the most effective one. First we calculated the force at which one must push the cart in order to move the cart effectively to understand how much stress would be put upon our design. Ultimately, we figured out that the required force would be anything from 108 Newtons all the way up to 317 Newtons of force, this led us to believe that the design must be:

1. closer to the ground where there is more turning power and the maximum amount of torque
2. Close proximity to the pre-existing handle for the most control

After analyzing the specifications that we would need we set to work building the prototype in this way. To test the device we set a control of 62 lbs plus a few boxes and tested the durability of the device at that point. We then had the client., Mr.Kiser, as well as several other students in the class attempt putting on and removing the device along with the testing of the usability of the design.

Our team tested several different variables when testing the design. We tested the stress, weight, and friction to ensure our design was not only durable, but comfortable and

easy to use. The stress continued to be tested as we tested the design daily to ensure that it worked in the same way as the day before. Although the weight of the cart never changed, we ensured that we our design would be able to function under upwards of 500 lbs. We did this using hot glue and pvc connectors to ensure the best stability and durability. Lastly the biggest variable that we took into consideration was the friction. This was a difficult one because we wanted the design to be able to lock into place but also have a removability factor.

Below is an image showing the prototype and the different variables that we measured above:



The one issue that arose when testing the second prototype was that the stress of pushing it caused the wood too much stress ultimately causing it to break. This led to us

retracing our steps and going back to the first prototype which worked better but needed modifications.

We were able to get several people to evaluate our design including the stakeholder as well as several different engineers from Vanderbilt along with a selective few classmates. Mr.Kiser's biggest worry was the focus on turning which seemed unreliable and quite shaky at some times. He commented, "turning is a little weird because it doesn't feel secure" and so we went to work to super glue the PVC together to ensure that the design would be able to withstand turning on a more realistic level. The Vandy engineers suggest two different things. Dr.London suggest that we, "create instructions for attachment," and "shorter the PVC across the bottom." Dr. Jared suggested that we figure out some way to better lock the device into the PVC whether that be a 3D print locking mechanism or some other way.

Ultimately we ended up taking these suggestions and incorporated them into the design. We ended up shortening the bottom PVC pipe that held the device together and then we added instructions to the device to show the user how specifically the cart should be used. Although Dr.Jared's idea was not directly answered, we ended up securing the PVC to the wood using hot glue because we were running short on time and didn't have the opportunity to add a more creative design solution.

Overall our design met several of the requirements that we set at the beginning of the design stages. We were able to incorporate several ideas that ultimately strengthened the design. The only criteria that is unable to be tested in its current state is the long-lasting

ability of the device; however, we have high expectations that this design will be durable enough to last far into the future if properly used. Moving forward we expect our design to continue to be tested and undergo further modifications because again, with the limited funding and time, we were unable to accomplish everything we wanted for the design. The next step would be the publicizing of the design to gain investors so that we can promote and improve our design.

Below are the three tested prototypes, each was put through a series of tests to test the several criterion:

